Name :J Mukhesh H.No:2303A51813 Batch:26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name:B. Tech	**Assignment Type: Lab**		**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s)Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju Mr. G. Kranthi		
Course Code	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/I	**Regulation**	R23
Date and Day of Assignment	Week 9 - Thursday	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All Batches
AssignmentNumber:18.4 (Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*ExpectedTime to complete*
1	**Lab 18: API Integration: Connecting to External Services with Error Handling**	Week 9

Lab Objectives

- Learn how to integrate Python programs with external REST APIs.
- Understand API request/response handling using requests or similar libraries.
- Implement proper error handling for failed API calls (timeouts, invalid responses, rate limits).
- Practice extracting and displaying meaningful information from API responses.

Learning Outcome

After completing this assignment, students will be able to:

- Integrate external REST APIs using AI-assisted Python scripting.
- Implement structured error handling for API failures and invalid inputs.
- Safely parse and validate JSON responses from external services.
- Develop resilient applications capable of handling real-world API limitations.

Task 1 – Movie Database API Integration

Scenario

You are building a movie recommendation tool that fetches movie details from a public Movie Database API.

Task

Use AI to generate a Python script that connects to a Movie API and retrieves movie information by title.

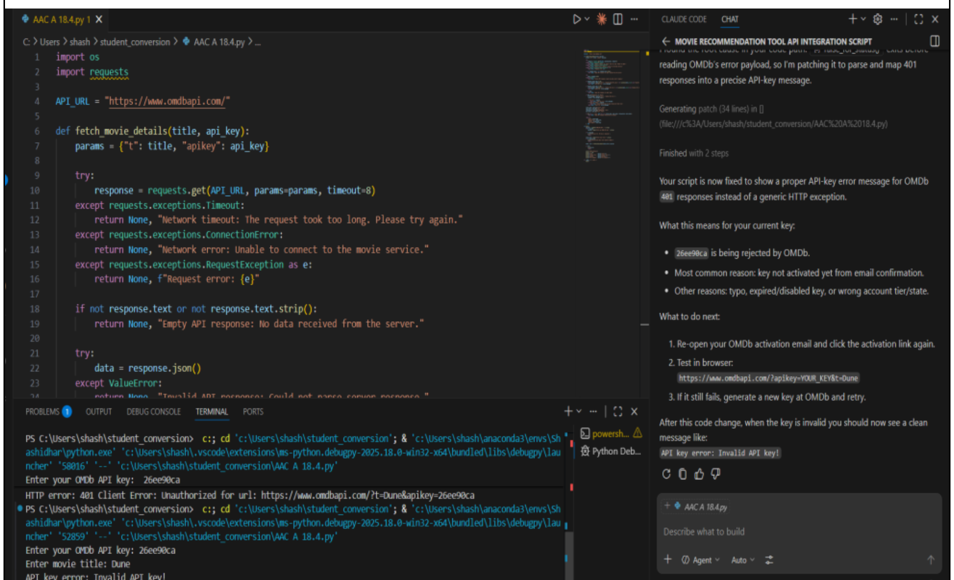
Instructions

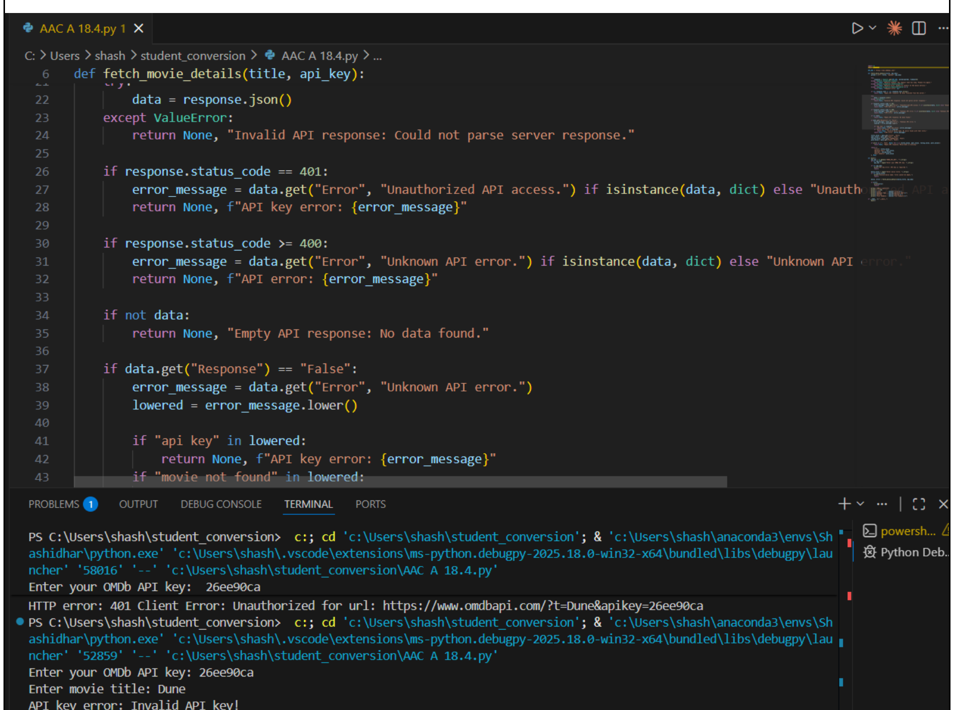
- Use `requests` library to fetch movie details.
- Accept movie title as user input.
- Display:
    - Title
    - Release year
    - Rating
    - Plot summary
- Implement error handling for:
    - Invalid movie name
    - API key errors

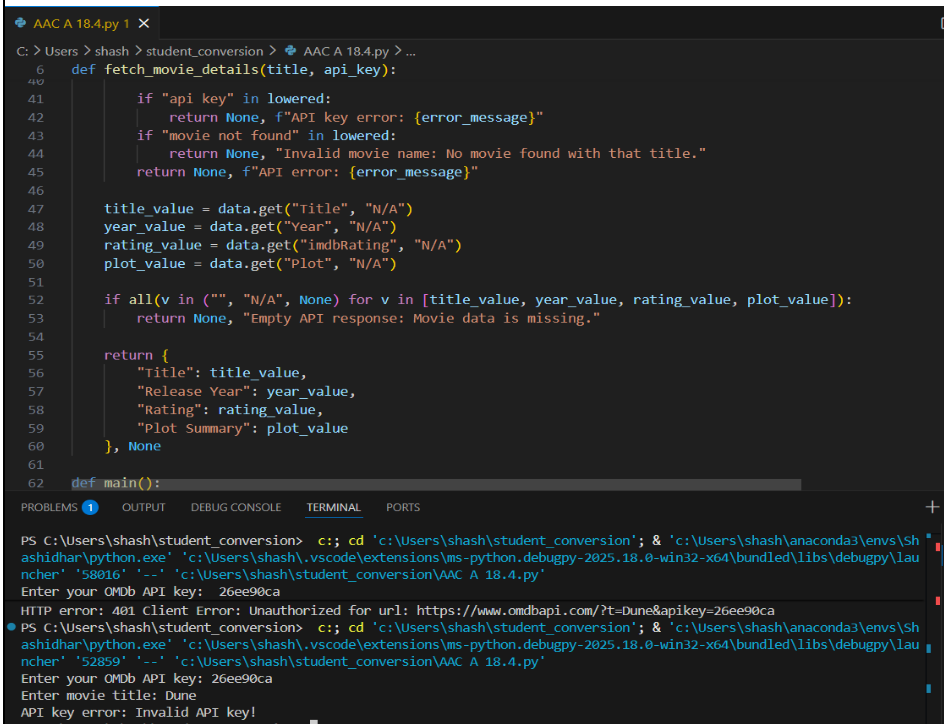
- Network timeout
- Empty API response

Expected Output

- A working Python script that fetches movie details.
- Graceful error messages instead of crashes.
- Clean, readable output format.







AAC A 18.4.py 1

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def fetch_movie_details(title, api_key):
            if "api key" in lowered:
                return None, f"API key error: {error_message}"
            if "movie not found" in lowered:
                return None, "Invalid movie name: No movie found with that title."
            return None, f"API error: {error_message}"

        title_value = data.get("Title", "N/A")
        year_value = data.get("Year", "N/A")
        rating_value = data.get("imdbRating", "N/A")
        plot_value = data.get("Plot", "N/A")

        if all(v in ("", "N/A", None) for v in [title_value, year_value, rating_value, plot_value]):
            return None, "Empty API response: Movie data is missing."

        return {
            "Title": title_value,
            "Release Year": year_value,
            "Rating": rating_value,
            "Plot Summary": plot_value
        }, None

def main():
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58016' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key:  26ee90ca
HTTP error: 401 Client Error: Unauthorized for url: https://www.omdbapi.com/?t=Dune&apikey=26ee90ca
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '52859' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
Enter movie title: Dune
API key error: Invalid API key!
```

```
AAC A 18.4.py 1 ×
C: > Users > shash > student_conversion > AAC A 18.4.py > ...
def main():
        return

    movie_title = input("Enter movie title: ").strip()
    if not movie_title:
        print("Invalid movie name: Title cannot be empty.")
        return

    movie, error = fetch_movie_details(movie_title, api_key)

    if error:
        print(error)
        return

    print("\nMovie Details")
    print("-" * 40)
    print(f"Title        : {movie['Title']}")
    print(f"Release Year : {movie['Release Year']}")
    print(f"Rating       : {movie['Rating']}")
    print(f"Plot Summary : {movie['Plot Summary']}")

if __name__ == "__main__":
    main()
```

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversi
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025
ncher' '58016' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key:  26ee90ca
HTTP error: 401 Client Error: Unauthorized for url: https://www.omdbapi.com/?t=
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversi
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025
ncher' '52859' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
Enter movie title: Dune
API key error: Invalid API key!
```

Task 2 – Public Transport Schedule API

Scenario

A city transport app needs to fetch bus/train arrival timings from a transport API.

Task

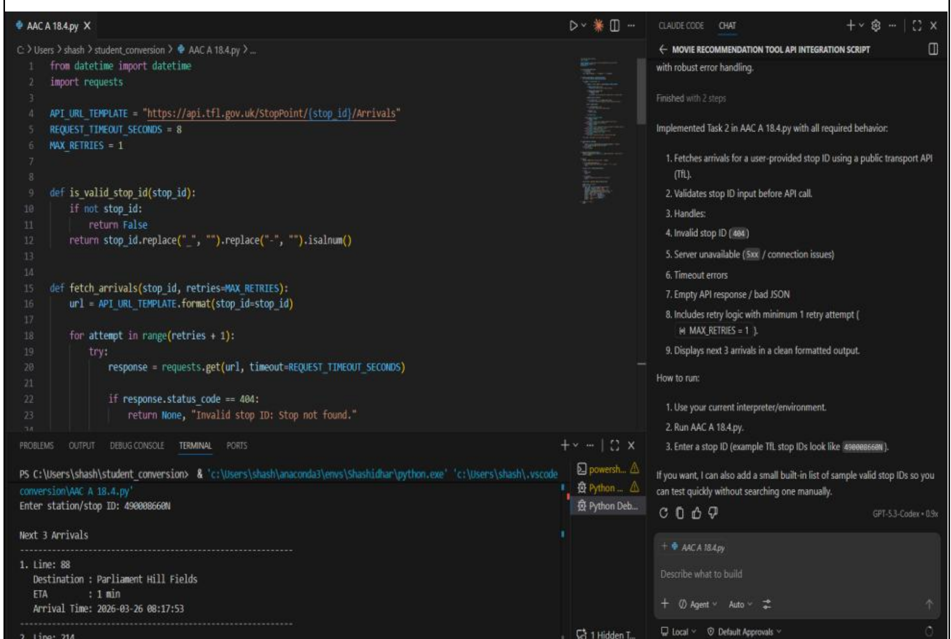
Use AI assistance to integrate a transport schedule API.

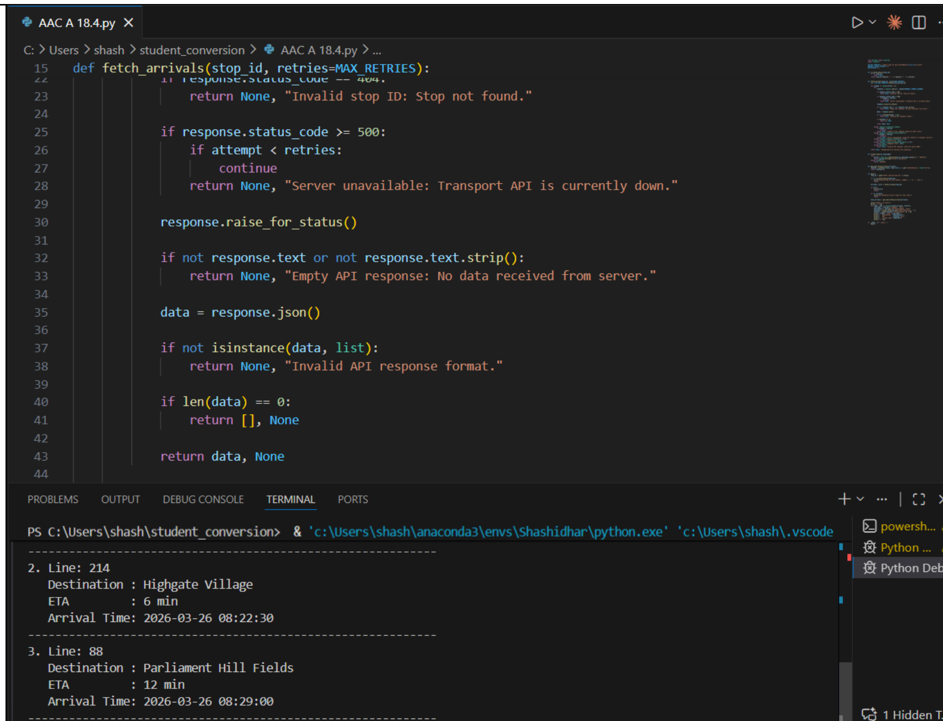
Instructions

- Fetch arrival times for a given station/stop ID.
- Validate user input.
- Handle errors such as:
  - Invalid stop ID
  - Server unavailable
  - Timeout errors
- Display next 3 arrival times in formatted output.
- Implement retry logic (minimum 1 retry attempt).

Expected Output

- Python script showing upcoming arrivals.
- Retry mechanism working properly.
- Proper exception handling using try-except blocks.





AAC A 18.4.py

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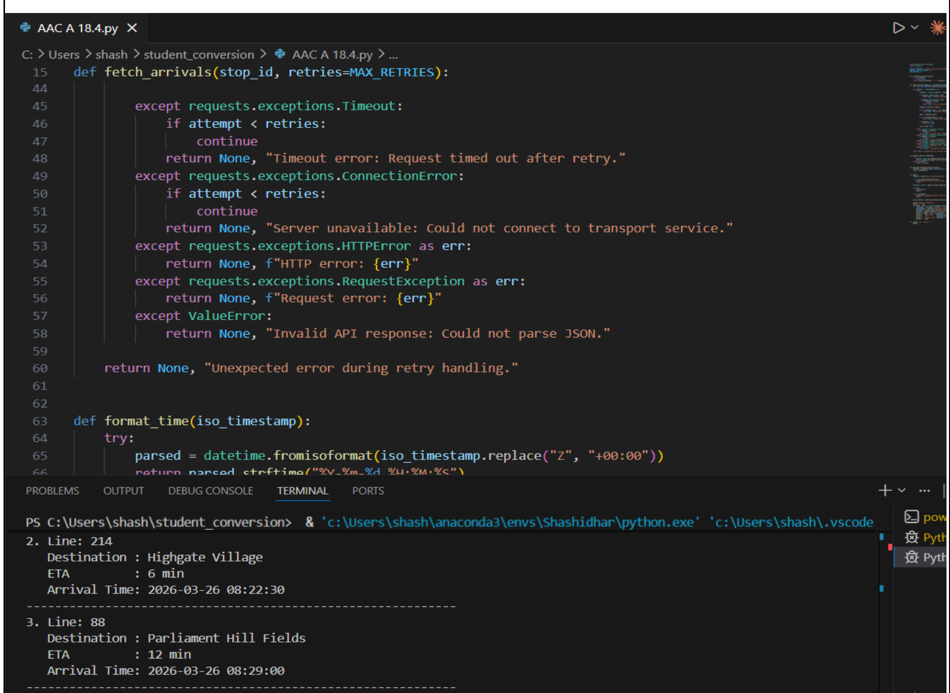
```python
def fetch_arrivals(stop_id, retries=MAX_RETRIES):
                return None, "Invalid stop ID: Stop not found."

            if response.status_code >= 500:
                if attempt < retries:
                    continue
                return None, "Server unavailable: Transport API is currently down."

            response.raise_for_status()

            if not response.text or not response.text.strip():
                return None, "Empty API response: No data received from server."

            data = response.json()

            if not isinstance(data, list):
                return None, "Invalid API response format."

            if len(data) == 0:
                return [], None

            return data, None
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode
------------------------------------------------------------
2. Line: 214
   Destination : Highgate Village
   ETA         : 6 min
   Arrival Time: 2026-03-26 08:22:30
------------------------------------------------------------
3. Line: 88
   Destination : Parliament Hill Fields
   ETA         : 12 min
   Arrival Time: 2026-03-26 08:29:00
------------------------------------------------------------
```



AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def fetch_arrivals(stop_id, retries=MAX_RETRIES):

        except requests.exceptions.Timeout:
            if attempt < retries:
                continue
            return None, "Timeout error: Request timed out after retry."
        except requests.exceptions.ConnectionError:
            if attempt < retries:
                continue
            return None, "Server unavailable: Could not connect to transport service."
        except requests.exceptions.HTTPError as err:
            return None, f"HTTP error: {err}"
        except requests.exceptions.RequestException as err:
            return None, f"Request error: {err}"
        except ValueError:
            return None, "Invalid API response: Could not parse JSON."

    return None, "Unexpected error during retry handling."


def format_time(iso_timestamp):
    try:
        parsed = datetime.fromisoformat(iso_timestamp.replace("Z", "+00:00"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode
2. Line: 214
   Destination : Highgate Village
   ETA         : 6 min
   Arrival Time: 2026-03-26 08:22:30
------------------------------------------------------------
3. Line: 88
   Destination : Parliament Hill Fields
   ETA         : 12 min
   Arrival Time: 2026-03-26 08:29:00
------------------------------------------------------------
```

AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def format_time(iso_timestamp):
        return "Unknown"


def get_next_three_arrivals(arrivals):
    ordered = sorted(arrivals, key=lambda x: x.get("timeToStation", float("inf")))
    return ordered[:3]


def main():
    stop_id = input("Enter station/stop ID: ").strip()

    if not is_valid_stop_id(stop_id):
        print("Invalid stop ID: Use letters, numbers, '-' or '_' only.")
        return

    arrivals, error = fetch_arrivals(stop_id)

    if error:
        print(error)
        return

    if not arrivals:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscod
2. Line: 214
   Destination : Highgate Village
   ETA         : 6 min
   Arrival Time: 2026-03-26 08:22:30
------------------------------------------------------------
3. Line: 88
   Destination : Parliament Hill Fields
   ETA         : 12 min
   Arrival Time: 2026-03-26 08:29:00
------------------------------------------------------------
```

AAC A 18.4.py ×

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def main():
    if not arrivals:
        print("No upcoming arrivals found for this stop.")
        return

    next_arrivals = get_next_three_arrivals(arrivals)

    print("\nNext 3 Arrivals")
    print("-" * 60)
    for index, item in enumerate(next_arrivals, start=1):
        line_name = item.get("lineName", "N/A")
        destination = item.get("destinationName", "N/A")
        expected = format_time(item.get("expectedArrival", ""))
        minutes = round(item.get("timeToStation", 0) / 60)
        print(f"{index}. Line: {line_name}")
        print(f"   Destination : {destination}")
        print(f"   ETA         : {minutes} min")
        print(f"   Arrival Time: {expected}")
        print("-" * 60)

if __name__ == "__main__":
    main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> & 'c:\Users\shash\anaconda3\envs\Shashidhar\pytho
2. Line: 214
   Destination : Highgate Village
   ETA         : 6 min
   Arrival Time: 2026-03-26 08:22:30
------------------------------------------------------------
3. Line: 88
   Destination : Parliament Hill Fields
   ETA         : 12 min
   Arrival Time: 2026-03-26 08:29:00
------------------------------------------------------------
```

Task 3 – Stock Market Data API

Scenario

You are developing a finance dashboard that retrieves live stock prices.

Task

Use AI to generate a Python script that fetches stock data from a financial API.

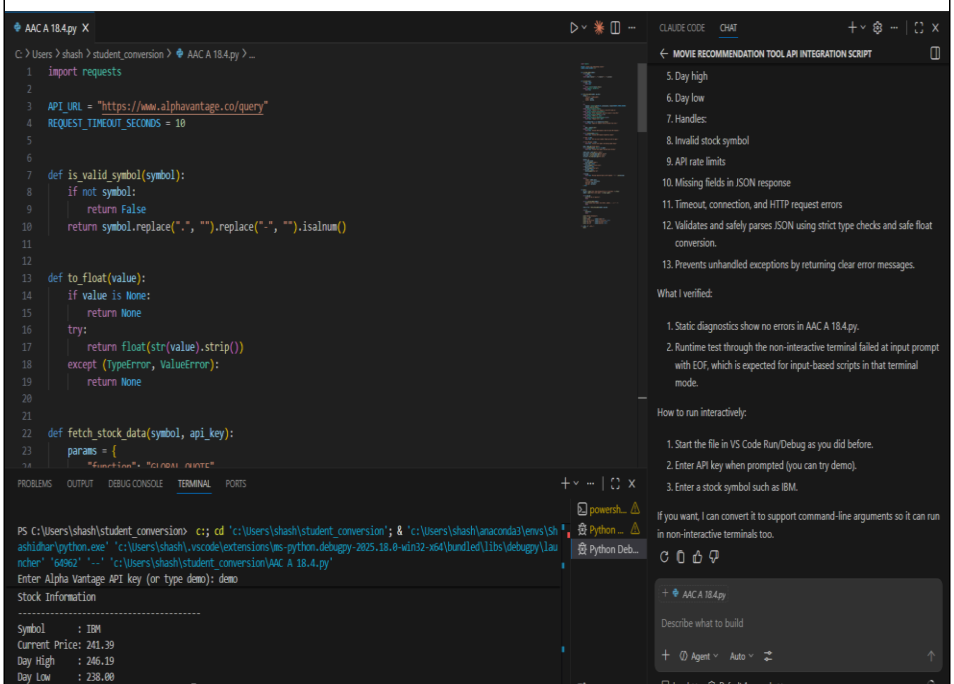
Instructions

- Accept stock symbol as input.

- Retrieve:
  - Current price
  - Day high
  - Day low
- Handle:
  - Invalid stock symbol
  - API rate limits
  - Missing fields in JSON response
- Validate and safely parse JSON response.

Expected Output

- Script fetches and displays stock information.
- No unhandled exceptions.
- Clear error messages for invalid symbols or rate limits.



AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def fetch_stock_data(symbol, api_key):
        "symbol": symbol,
        "apikey": api_key,
    }

    try:
        response = requests.get(API_URL, params=params, timeout=REQUEST_TIMEOUT_SECONDS)
        response.raise_for_status()
    except requests.exceptions.Timeout:
        return None, "Timeout error: Request to stock API timed out."
    except requests.exceptions.ConnectionError:
        return None, "Server unavailable: Could not connect to stock API."
    except requests.exceptions.HTTPError as err:
        return None, f"HTTP error: {err}"
    except requests.exceptions.RequestException as err:
        return None, f"Request error: {err}"

    if not response.text or not response.text.strip():
        return None, "Empty API response: No data received from server."

    try:
        data = response.json()
    except ValueError:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anac
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\lib
ncher' '64962' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter Alpha Vantage API key (or type demo): demo
Stock Information
----------------------------------------
Symbol       : IBM
Current Price: 241.39
Day High     : 246.19
Day Low      : 238.00
PS C:\Users\shash\student_conversion>
```

AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def fetch_stock_data(symbol, api_key):
        except ValueError:
            return None, "Invalid JSON response: Could not parse API response."

        if not isinstance(data, dict):
            return None, "Invalid JSON response: Expected an object."

        if "Note" in data:
            return None, "API rate limit reached: Please wait and try again."

        if "Error Message" in data:
            return None, "Invalid stock symbol: No matching symbol found."

        quote = data.get("Global Quote")
        if not isinstance(quote, dict) or not quote:
            return None, "Invalid stock symbol: No quote data returned."

        symbol_value = quote.get("01. symbol")
        price_value = to_float(quote.get("05. price"))
        high_value = to_float(quote.get("03. high"))
        low_value = to_float(quote.get("04. low"))

        missing = []
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\U
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-
ncher' '64962' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter Alpha Vantage API key (or type demo): demo
Stock Information
----------------------------------------
Symbol       : IBM
Current Price: 241.39
Day High     : 246.19
Day Low      : 238.00
PS C:\Users\shash\student_conversion>
```

AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def main():
        print("API key is required.")
        return

    if not is_valid_symbol(symbol):
        print("Invalid stock symbol: Use letters, numbers, '.', or '-'.")
        return

    stock, error = fetch_stock_data(symbol, api_key)

    if error:
        print(error)
        return

    print("\nStock Information")
    print("-" * 40)
    print(f"Symbol       : {stock['symbol']}")
    print(f"Current Price: {stock['current_price']:.2f}")
    print(f"Day High     : {stock['day_high']:.2f}")
    print(f"Day Low      : {stock['day_low']:.2f}")

if __name__ == "__main__":
    main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Us
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x
ncher' '64962' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter Alpha Vantage API key (or type demo): demo
Stock Information
----------------------------------------
Symbol       : IBM
Current Price: 241.39
Day High     : 246.19
Day Low      : 238.00
PS C:\Users\shash\student_conversion>
```

Task 4 – Geolocation API (IP Address Lookup)

Scenario

A cybersecurity tool needs to detect the geographical location of a user based on IP address.

Task

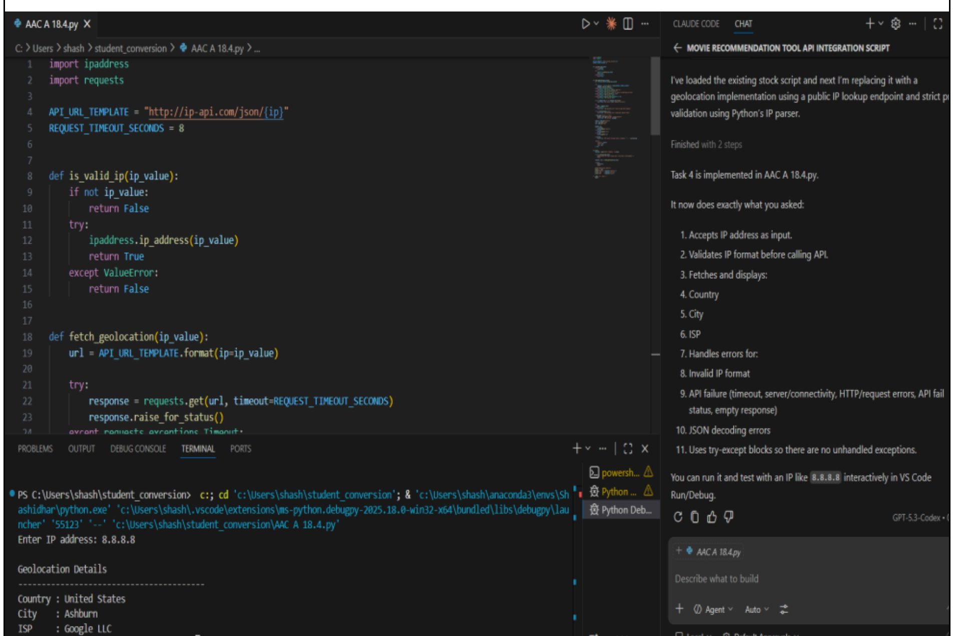
Integrate a Geolocation API using AI assistance.

Instructions

- Accept IP address as input.
- Fetch:
  - Country
  - City
  - ISP
- Validate IP format before making API call.
- Handle:
  - Invalid IP format
  - API failure
  - JSON decoding errors

Expected Output

- A script that returns geolocation details.
- Input validation implemented.
- Robust error handling included.



AAC A 18.4.py ×

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def fetch_geolocation(ip_value):
        except requests.exceptions.Timeout:
            return None, "API failure: Request timed out."
        except requests.exceptions.ConnectionError:
            return None, "API failure: Could not connect to geolocation service."
        except requests.exceptions.HTTPError as err:
            return None, f"API failure: {err}"
        except requests.exceptions.RequestException as err:
            return None, f"API failure: {err}"

        if not response.text or not response.text.strip():
            return None, "API failure: Empty response from server."

        try:
            data = response.json()
        except ValueError:
            return None, "JSON decoding error: Could not parse API response."

        if not isinstance(data, dict):
            return None, "JSON decoding error: Unexpected response format."

        if data.get("status") == "fail":
            message = data.get("message", "Lookup failed.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shas
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundl
ncher' '55123' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter IP address: 8.8.8.8

Geolocation Details
----------------------------------------
Country : United States
City    : Ashburn
ISP     : Google LLC
PS C:\Users\shash\student_conversion>
```

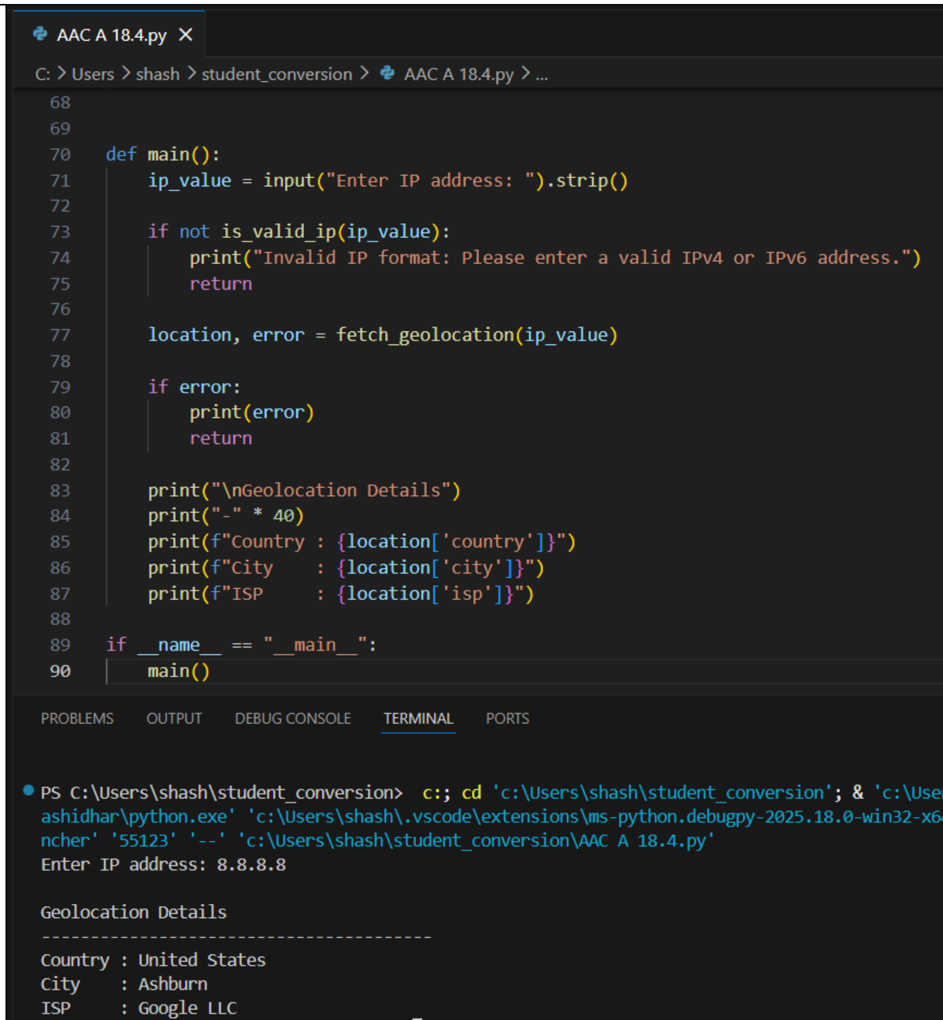
AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python
def fetch_geolocation(ip_value):
    if data.get("status") == "fail":
        message = data.get("message", "Lookup failed.")
        return None, f"API failure: {message}"

    country = data.get("country")
    city = data.get("city")
    isp = data.get("isp")

    missing = []
    if not country:
        missing.append("country")
    if not city:
        missing.append("city")
    if not isp:
        missing.append("isp")

    if missing:
        return None, "API failure: Missing fields in response: " + ", ".join(missing)

    return {
        "country": country,
        "city": city,
        "isp": isp,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaco
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs
ncher' '55123' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter IP address: 8.8.8.8

Geolocation Details
----------------------------------------
Country : United States
City    : Ashburn
ISP     : Google LLC
```



AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```python


def main():
    ip_value = input("Enter IP address: ").strip()

    if not is_valid_ip(ip_value):
        print("Invalid IP format: Please enter a valid IPv4 or IPv6 address.")
        return

    location, error = fetch_geolocation(ip_value)

    if error:
        print(error)
        return

    print("\nGeolocation Details")
    print("-" * 40)
    print(f"Country : {location['country']}")
    print(f"City    : {location['city']}")
    print(f"ISP     : {location['isp']}")

if __name__ == "__main__":
    main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\User
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64
ncher' '55123' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter IP address: 8.8.8.8

Geolocation Details
----------------------------------------
Country : United States
City    : Ashburn
ISP     : Google LLC
```

Task 5 – Real-Time Application: Payment Gateway Simulation API

Scenario

You are building a test payment integration system that connects to a sandbox payment API.

Task

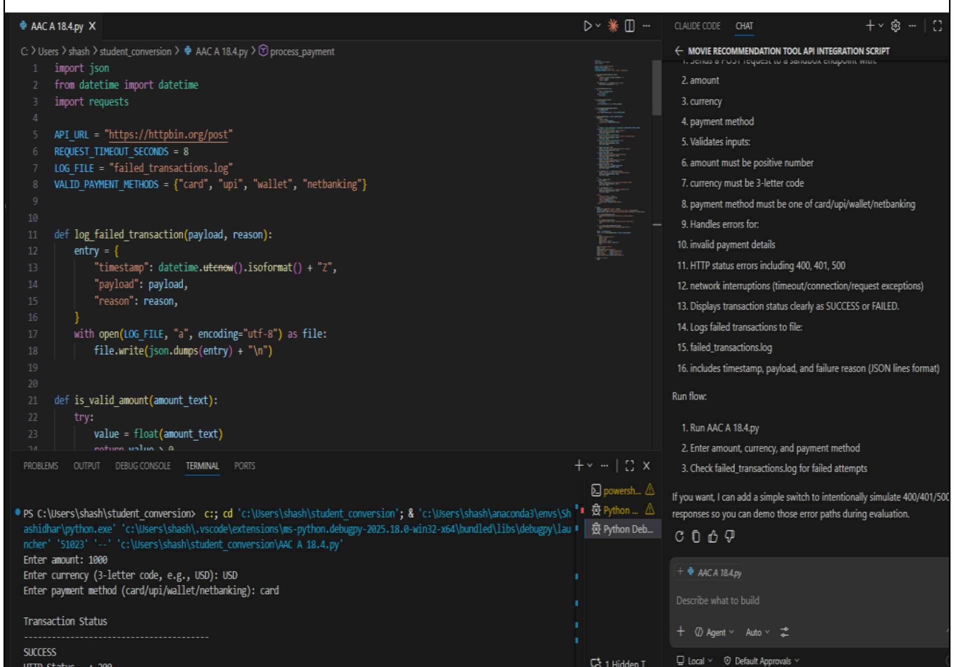
Use AI to generate a Python script that simulates payment processing.

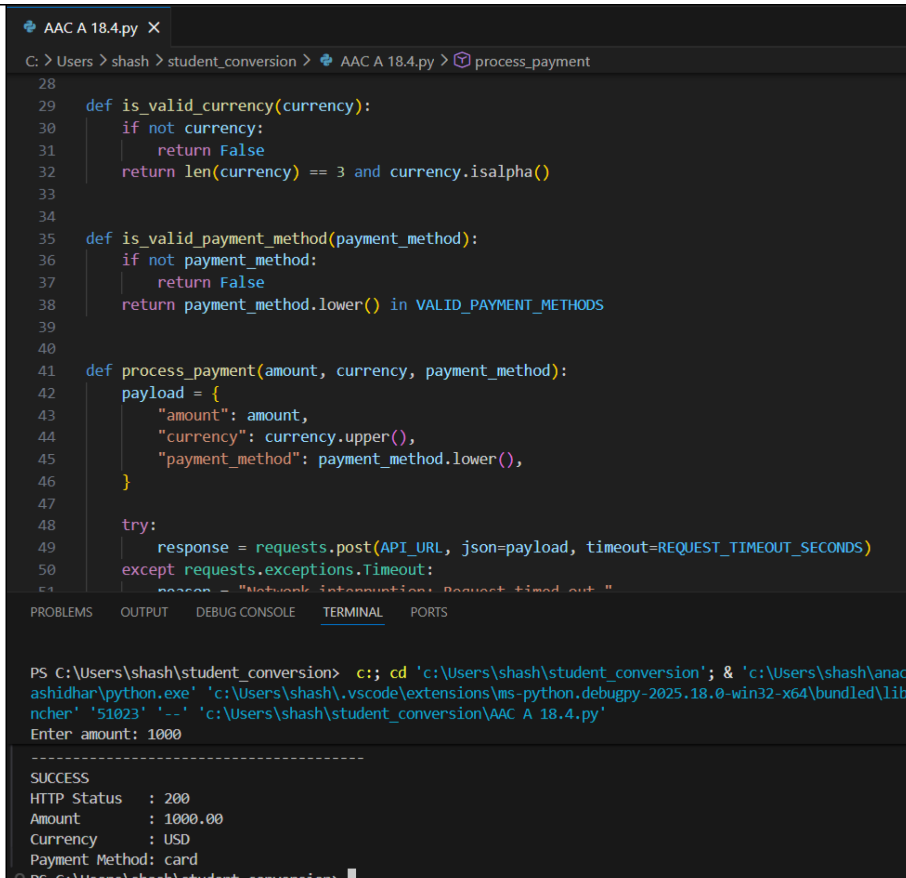
Instructions

- Send POST request with:
  - Amount
  - Currency
  - Payment method
- Validate input values.
- Handle:
  - Invalid payment details
  - HTTP status errors (400, 401, 500)
  - Network interruptions
- Display transaction status clearly.
- Log failed transactions to a file.

Expected Output

- Script processes simulated payment.
- Errors handled gracefully.
- Failed requests logged properly.






```python
def is_valid_currency(currency):
    if not currency:
        return False
    return len(currency) == 3 and currency.isalpha()


def is_valid_payment_method(payment_method):
    if not payment_method:
        return False
    return payment_method.lower() in VALID_PAYMENT_METHODS


def process_payment(amount, currency, payment_method):
    payload = {
        "amount": amount,
        "currency": currency.upper(),
        "payment_method": payment_method.lower(),
    }

    try:
        response = requests.post(API_URL, json=payload, timeout=REQUEST_TIMEOUT_SECONDS)
    except requests.exceptions.Timeout:
```



```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anac
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
----------------------------------------
SUCCESS
HTTP Status   : 200
Amount        : 1000.00
Currency      : USD
Payment Method: card
PS C:\Users\shash\student_conversion>
```

AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```python
def process_payment(amount, currency, payment_method):
            reason = "Network interruption: Request timed out."
            log_failed_transaction(payload, reason)
            return None, reason
        except requests.exceptions.ConnectionError:
            reason = "Network interruption: Unable to connect to payment gateway."
            log_failed_transaction(payload, reason)
            return None, reason
        except requests.exceptions.RequestException as err:
            reason = f"Network interruption: {err}"
            log_failed_transaction(payload, reason)
            return None, reason

        if response.status_code == 400:
            reason = "HTTP 400: Invalid payment details sent to gateway."
            log_failed_transaction(payload, reason)
            return None, reason
        if response.status_code == 401:
            reason = "HTTP 401: Unauthorized request to gateway."
            log_failed_transaction(payload, reason)
            return None, reason
        if response.status_code == 500:
            reason = "HTTP 500: Payment gateway server error."
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
----------------------------------------
SUCCESS
HTTP Status   : 200
Amount        : 1000.00
Currency      : USD
Payment Method: card
```

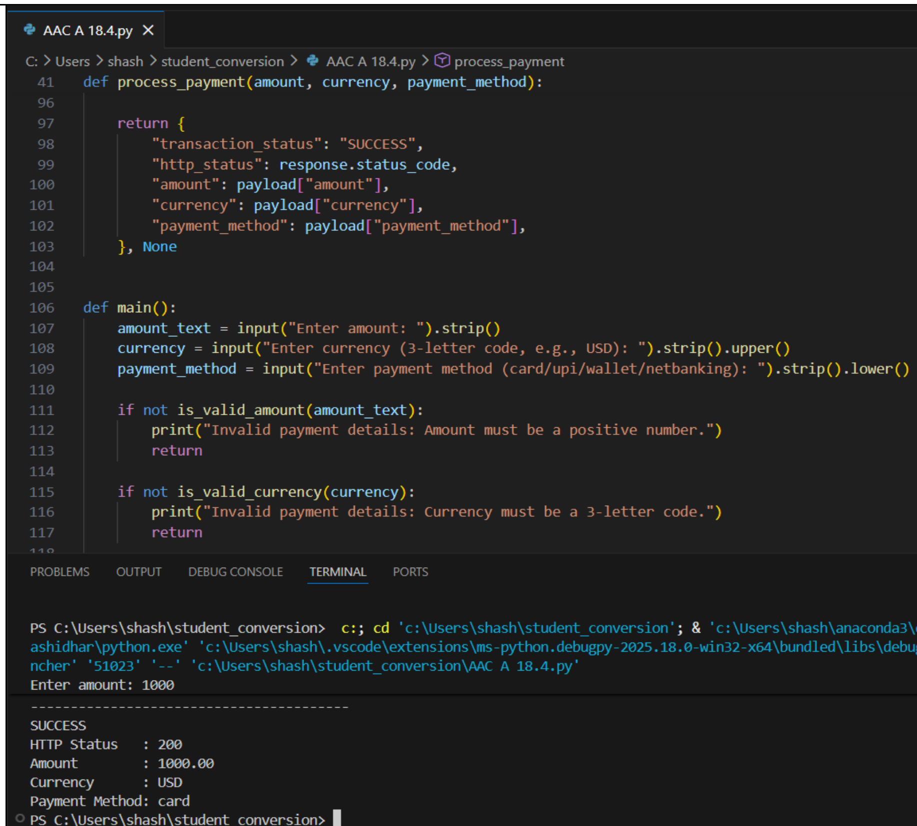
AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```python
def process_payment(amount, currency, payment_method):
            return None, reason
        if response.status_code >= 400:
            reason = f"HTTP error: {response.status_code}"
            log_failed_transaction(payload, reason)
            return None, reason

        if not response.text or not response.text.strip():
            reason = "API failure: Empty response from payment gateway."
            log_failed_transaction(payload, reason)
            return None, reason

        try:
            data = response.json()
        except ValueError:
            reason = "JSON decoding error: Could not parse gateway response."
            log_failed_transaction(payload, reason)
            return None, reason

        if not isinstance(data, dict):
            reason = "JSON decoding error: Unexpected response format."
            log_failed_transaction(payload, reason)
            return None, reason
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
----------------------------------------
SUCCESS
HTTP Status   : 200
Amount        : 1000.00
Currency      : USD
Payment Method: card
PS C:\Users\shash\student_conversion>
```



```
AAC A 18.4.py
C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment
def process_payment(amount, currency, payment_method):

    return {
        "transaction_status": "SUCCESS",
        "http_status": response.status_code,
        "amount": payload["amount"],
        "currency": payload["currency"],
        "payment_method": payload["payment_method"],
    }, None


def main():
    amount_text = input("Enter amount: ").strip()
    currency = input("Enter currency (3-letter code, e.g., USD): ").strip().upper()
    payment_method = input("Enter payment method (card/upi/wallet/netbanking): ").strip().lower()

    if not is_valid_amount(amount_text):
        print("Invalid payment details: Amount must be a positive number.")
        return

    if not is_valid_currency(currency):
        print("Invalid payment details: Currency must be a 3-letter code.")
        return
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\e
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debug
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
----------------------------------------
SUCCESS
HTTP Status    : 200
Amount         : 1000.00
Currency       : USD
Payment Method: card
PS C:\Users\shash\student_conversion>
```

AAC A 18.4.py

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```python
def main():

        amount = float(amount_text)
        result, error = process_payment(amount, currency, payment_method)

        if error:
            print("\nTransaction Status")
            print("-" * 40)
            print("FAILED")
            print(f"Reason: {error}")
            print(f"Failure logged to {LOG_FILE}")
            return

        print("\nTransaction Status")
        print("-" * 40)
        print(result["transaction_status"])
        print(f"HTTP Status   : {result['http_status']}")
        print(f"Amount        : {result['amount']:.2f}")
        print(f"Currency      : {result['currency']}")
        print(f"Payment Method: {result['payment_method']}")

if __name__ == "__main__":
    main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion';
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
----------------------------------------
SUCCESS
HTTP Status   : 200
Amount        : 1000.00
Currency      : USD
Payment Method: card
```

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots